

PROJECT REPORT

150 kW DC Fast-Charging Station Gadag, Karnataka

Site survey · technical design · project cost · demand & revenue projection

Prepared for	Mr. Manjunath
Prepared by	Sonar.EV
Site	Hubli–Gadag Road (NH-67), Binkadakatti, Gadag taluk · 15.42335° N, 75.57571° E
Charger OEM	Exicom (1 × 120 kW + 1 × 30 kW, dual-gun DC)
Date	8 October 2026

1. Executive summary

Mr. Manjunath's plot on the Hubli–Gadag stretch of NH-67 can host a 150 kW DC fast-charging station: one 120 kW and one 30 kW dual-gun Exicom charger (four vehicles at a time) under a canopy, fed by a new 200 kVA 11 kV substation. Total project cost is ₹49,14,798 before GST (₹51,60,538 with 5% GST).

₹49.15 L Project cost ex-GST	150 kW 120 kW + 30 kW, 4 guns	₹51 L Revenue in year 5 (base)	~5.5 yrs Base-case payback
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- **Location:** a roadside plot with 60 ft of frontage on a four-lane trunk road, about 6 km west of Gadag town on the route to Hubballi.
- **Demand:** no published traffic count exists for this stretch, so the projection assumes 12,000 vehicles a day; a 7-day count should confirm it before work orders are signed.
- **Returns:** the base case turns EBITDA-positive in year 3 and recovers the investment in about 5.5 years. The conservative case does not pay back within 7 years; the optimistic case pays back in 3.8 years.
- **Biggest lever:** the share of passing cars that are electric. Karnataka's electric-car sales roughly doubled in FY 2025-26, which supports a rising curve, but Gadag district is still early.

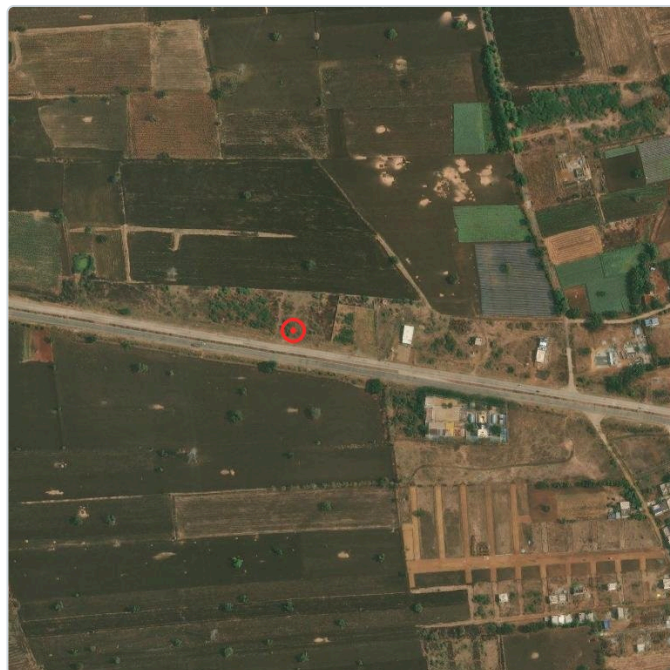
Read this as a planning estimate. Costs come from The Lynx quotation (6 Oct 2026) and the charger and canopy prices you provided. Demand and revenue rest on the assumptions in section 6 and are not guaranteed.

2. Location & satellite survey

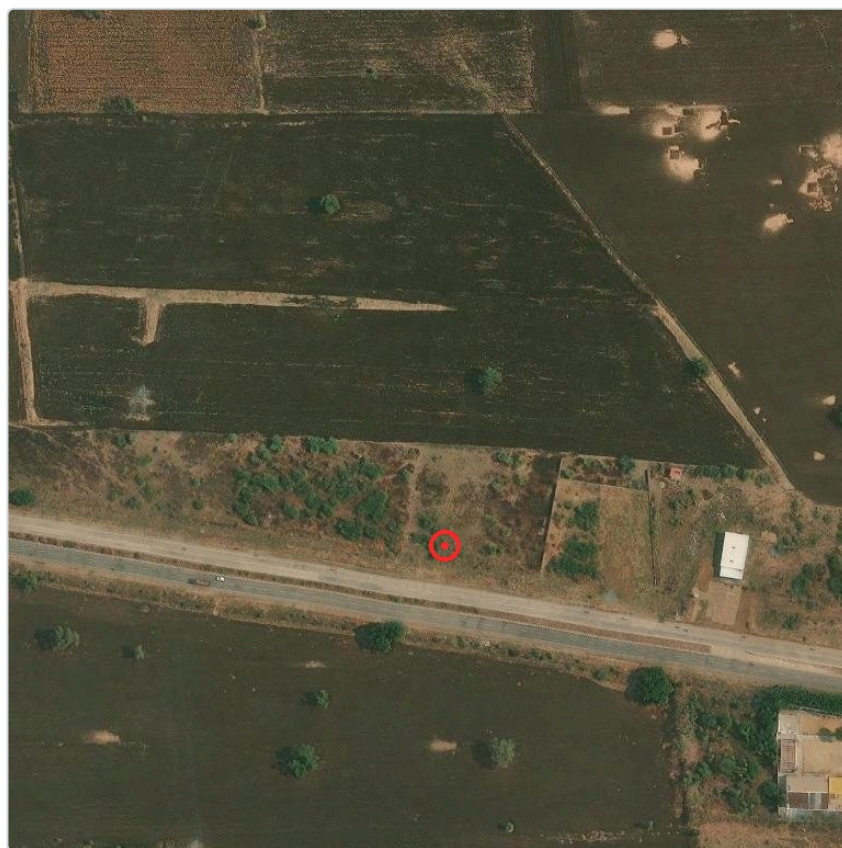
The site sits on the north side of the Hubli–Gadag Road, a trunk road classified as NH-67, in Binkadakatti village, Gadag taluk. Gadag town is about 6 km east; Hubballi is about 50 km west. Open fields lie on both sides, so there is room for a substation yard and a large canopy.



Regional view (~3.5 km across). Red ring = site.



Corridor view (~880 m across). The road runs east–west just south of the marker.



Close view (~440 m across): plot frontage on the road, open land behind, a small building and layout plots to the east.

Imagery: Esri World Imagery (© Esri, Maxar, Earthstar Geographics and the GIS user community). Retrieved 8 Oct 2026; capture dates vary.

3. Site survey & layout

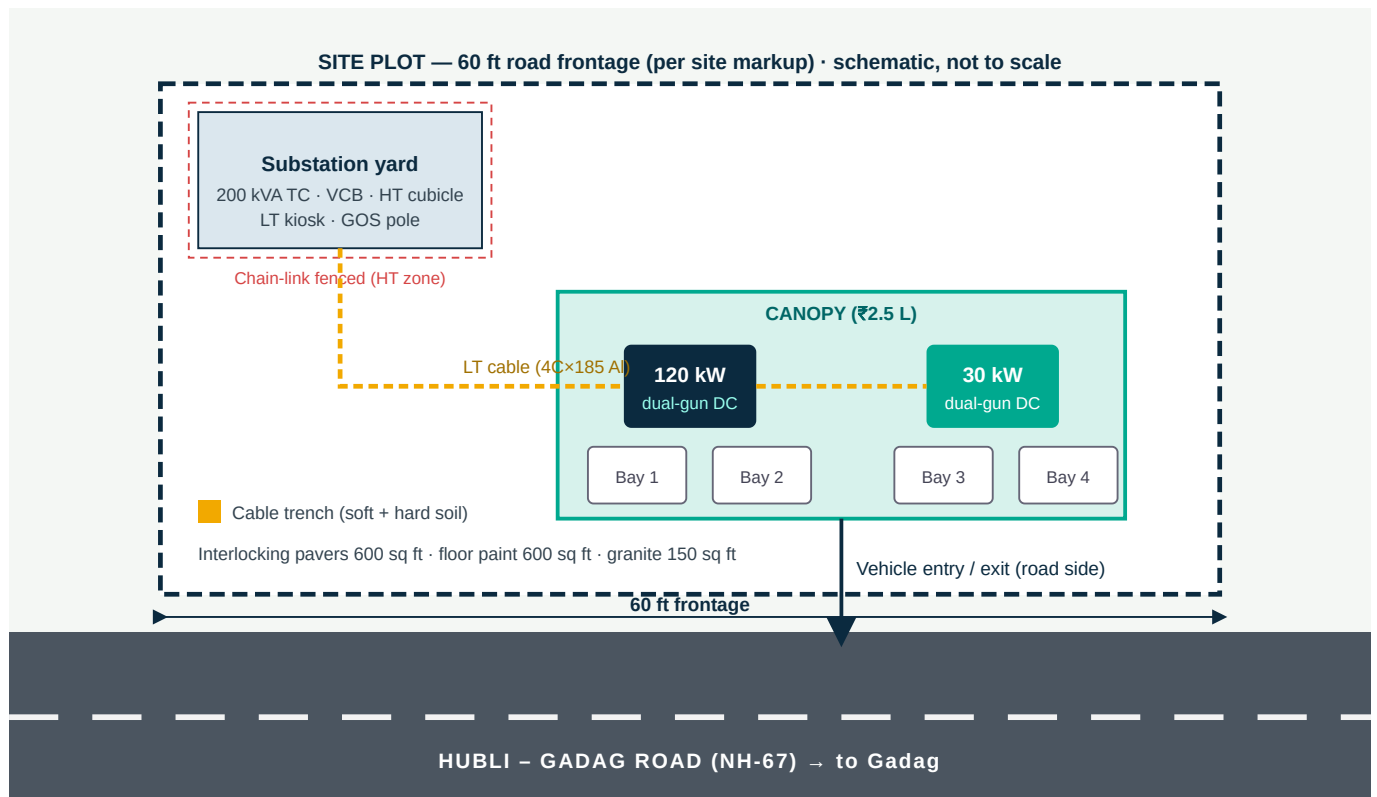


Site markup supplied by the client: 60 ft of road frontage (Google Maps, scale bar 50 ft / 20 m).

Findings

- Road frontage: **60 ft**, taken from the markup. Plot depth and boundary sketch still need an on-site tape measurement.
- Direct road access with no visible obstruction; entry and exit can be on the highway side.
- Flat, open, lightly vegetated land; hard soil is allowed for in the trenching rate.
- HT supply must be extended to the plot; 44 m of 11 kV cable and a pole-mounted GOS are in the quotation.
- Fire kit, bollards, car stoppers and signage are included for safety.

Site-visit items to close before ordering: depth of plot, nearest HT pole and its distance, soil test, title and road-access permission from NHA/PWD for the entry cut.

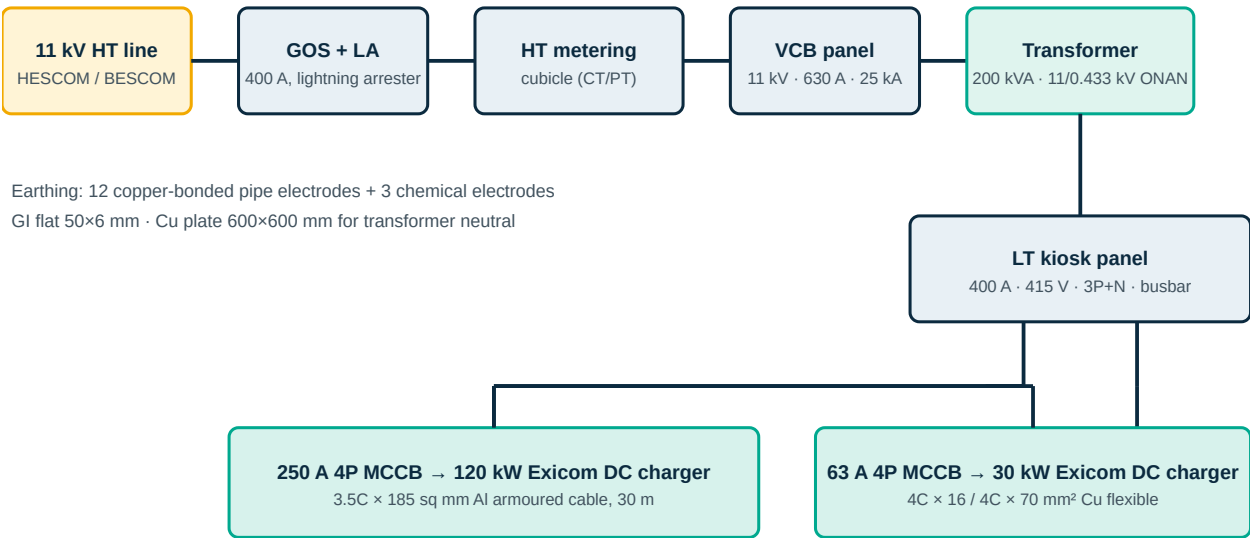


Schematic layout. Four parking bays under one canopy, substation yard at the rear, cable trench between them. Positions are indicative.

4. Technical design

Item	Specification
Chargers	Exicom 120 kW dual-gun DC fast charger (1 No.) and Exicom 30 kW dual-gun DC charger (1 No.)
Vehicles served	4 at once (2 guns per charger; power is shared between guns on each unit)
Supply	11 kV HT, 200 kVA ONAN transformer 11/0.433 kV, VCB 630 A / 25 kA, HT metering cubicle
LT distribution	400 A, 415 V LT kiosk with 250 A 4P MCCB (120 kW) and 63 A 4P MCCB (30 kW)
Earthing	12 copper-bonded pipe electrodes, 3 chemical electrodes, GI flat and copper strip, copper plate for transformer neutral
Safety	Fire extinguishers and buckets, danger boards, shock chart, cable markers, bollards, car stoppers, chain-link fence
Civil	RCC charger foundation 4700×1000×400 mm, equipment foundations, interlocking pavers 600 sq ft, painted bays 600 sq ft

Single-line power flow: 11 kV supply to the two chargers

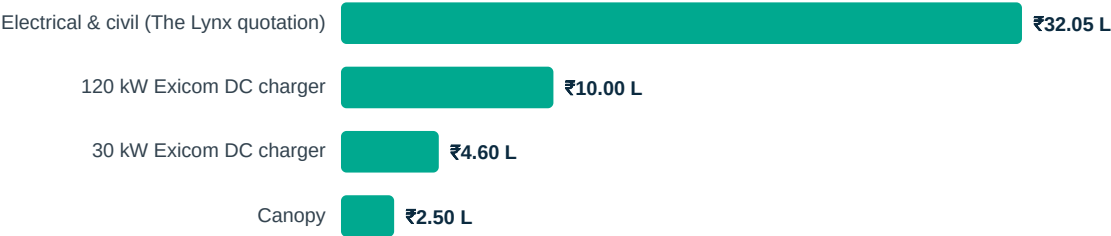


Single-line power flow. The 200 kVA transformer covers both chargers (150 kW nameplate) with headroom for auxiliaries.

The quotation names BESCOM for sanction. Gadag district is normally served by HESCOM; please confirm the correct distribution company before applying.

5. Project cost

Project cost split (₹, ex-GST)

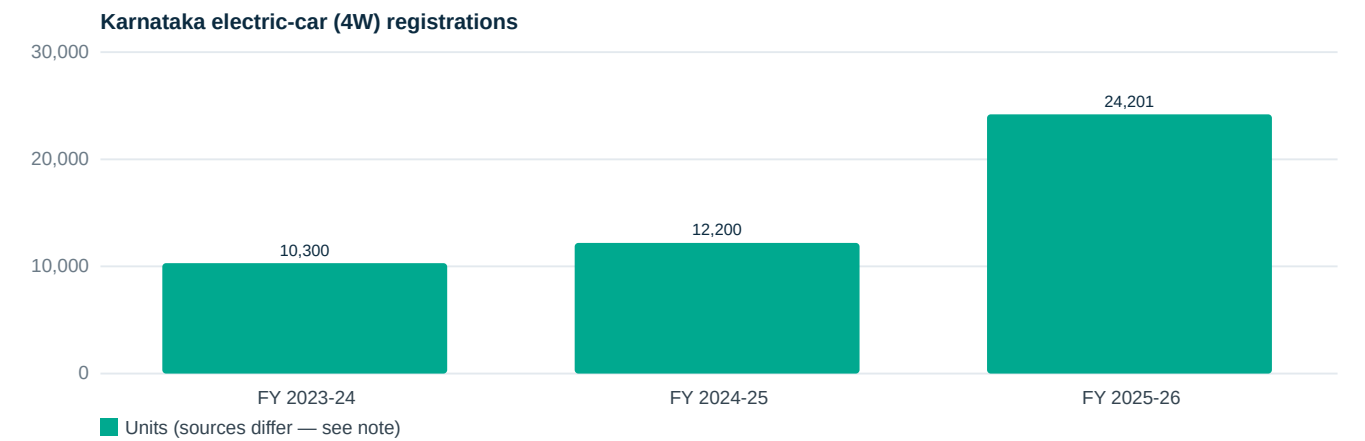


Component	Amount (₹)	Source
Electrical & civil infrastructure (59 items)	32,04,798	The Lynx quotation, 6 Oct 2026
Canopy	2,50,000	Client-provided
30 kW Exicom dual-gun DC charger	4,60,000	Client-provided
120 kW Exicom dual-gun DC charger	10,00,000	Client-provided
Total before GST	49,14,798	
GST @ 5%	2,45,739.90	Client-confirmed rate (quotation said 18%)
Total with GST	51,60,537.90	

Not included: statutory deposits and inspection fees (CEIG, CT/PT testing, TAQC/MRT, security deposit), which the quotation lists "at actuals". Budget for these separately. GST paid on capex may be claimable as input credit if the owner is GST-registered; confirm with a tax adviser.

6. Market & demand analysis

Karnataka is India's second-largest electric-car market. Registrations rose from about 10,300 (FY 2023-24) to 12,200 (FY 2024-25) and then to 24,201 in FY 2025-26, about 11% of India's electric-car sales. Nationally, electric vehicles were 8% of all new registrations in calendar 2025.



FY 2023-24 and 2024-25 figures: [Deccan Herald](#). FY 2025-26: [EVreporter \(Vahan data\)](#), which notes some states' data are partial. Sources differ in scope, so treat the growth as indicative. National share: [All India Radio](#).

Competition and corridor

- A public directory lists only two charging points in Gadag (an Ather station at Bellad Hyundai on Hubli Road and an IOCL station on Mulgund Road), with no DC fast charger named. See [91trucks](#). Verify on site.
- NH-67 links Dharwad, Hubballi, Gadag, Koppal, Hosapete and Ballari, and carries traffic to Hampi. Source: [Wikipedia](#). A Hubballi–Hosapete four-laning project passes through Gadag.

How many EVs pass the site? (estimate)

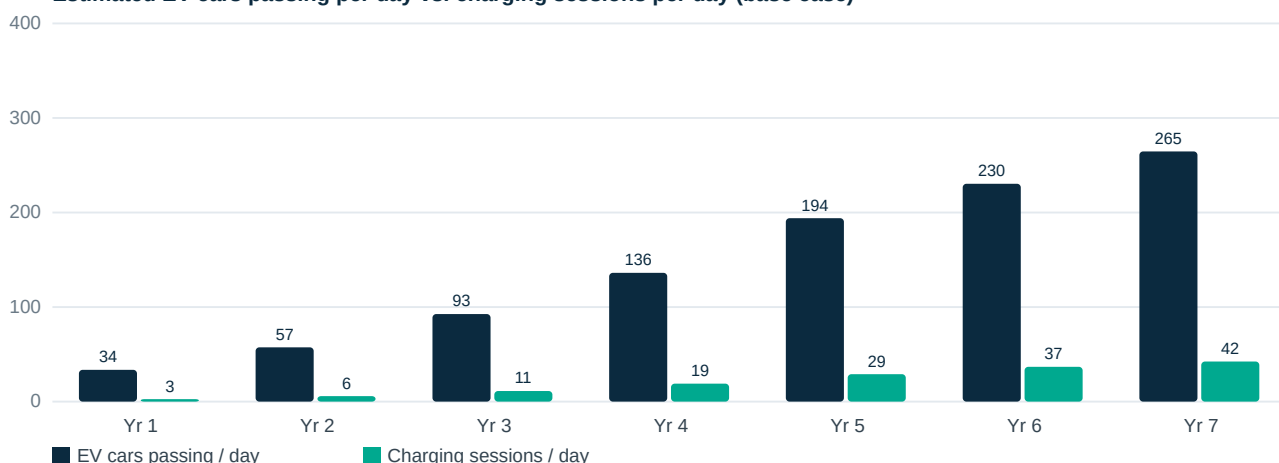
No published vehicle count exists for this stretch, so the estimate below is built from stated assumptions. Replace them with a real 7-day count.

Assumption	Base value	Basis
Vehicles per day at site, year 1	12,000, growing 5% a year	Assumed for a four-lane trunk road between two district towns; needs a traffic count
Cars (4W) in the traffic mix	35%	Typical for Indian national highways
Electric share of cars	0.8% in year 1, rising to 4.7% by year 7	Karnataka e-car sales growth; fleet share lags sales share
Share of passing EVs that stop to charge	8% rising to 16%	Rises with awareness and as no other fast charger is nearby
Energy per session	24 kWh	Typical top-up on a highway trip
Selling price	₹20 per kWh, before GST	Typical public DC tariff; set by the owner
Power cost	₹7 per kWh, all-in	Assumed HT tariff plus losses; confirm with the distribution company
Running cost	₹7.2 L a year, +5% a year	Two attendants, maintenance contract, insurance, connectivity. Land assumed owned
Payment fees	3% of revenue	Gateway and charge-point software

7. Revenue projection & returns

In the base case the station serves about 3 cars a day in year 1 and 29 a day in year 5. Revenue reaches ₹51.0 lakh in year 5 and the cumulative cash position turns positive in about 5.5 years.

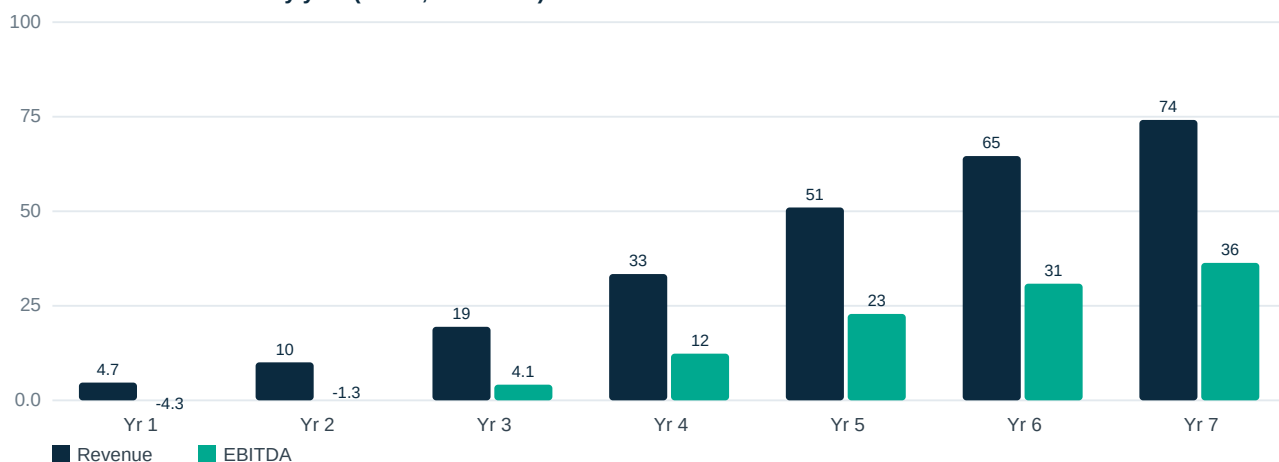
Estimated EV cars passing per day vs. charging sessions per day (base case)



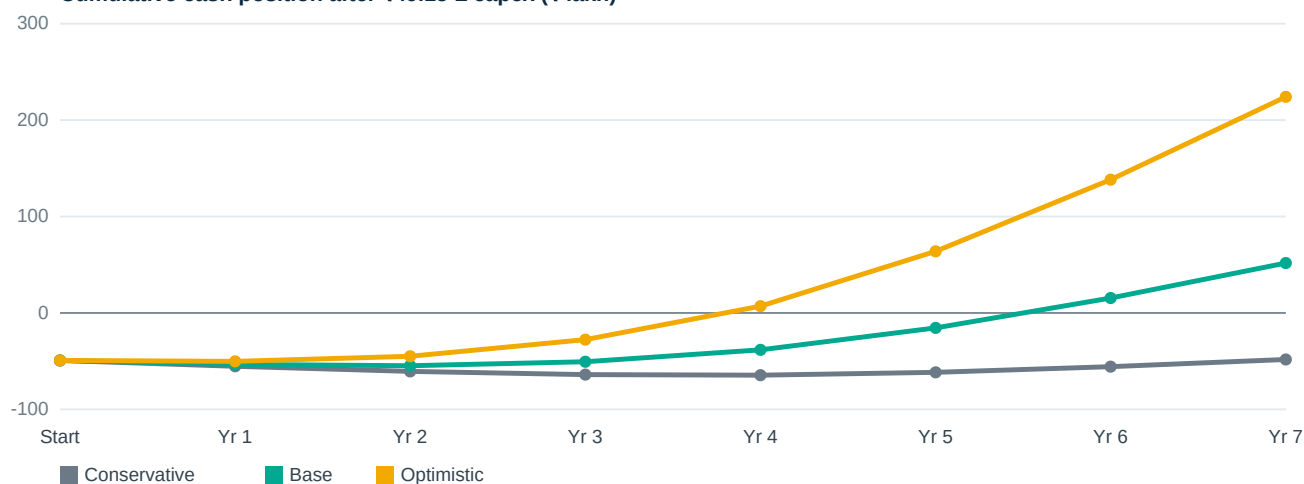
Year	Vehicles / day	EV cars passing	Sessions / day	kWh / day	Use*	Revenue (₹)	EBITDA (₹)	Cumulative (₹)
Year 1	12,000	34	2.7	65	2%	4,70,938	-4,28,019	-53,42,817
Year 2	12,600	57	5.7	138	4%	10,04,422	-1,33,259	-54,76,075
Year 3	13,230	93	11.1	267	7%	19,47,033	4,13,360	-50,62,715
Year 4	13,892	136	19.1	457	13%	33,39,161	12,36,790	-38,25,925
Year 5	14,586	194	29.1	698	19%	50,98,183	22,85,709	-15,40,216
Year 6	15,315	230	36.9	885	25%	64,61,276	30,87,069	15,46,853
Year 7	16,081	265	42.3	1,016	28%	74,15,442	36,32,705	51,79,558

*Use = energy sold ÷ (150 kW × 24 h). EBITDA = revenue – power – payment fees – running costs; tax and depreciation excluded. Cumulative starts at –₹49,14,798 (capex before GST).

Revenue vs. EBITDA by year (₹ lakh, base case)



Cumulative cash position after ₹49.15 L capex (₹ lakh)



Scenarios

Scenario	Assumptions	Year-5 revenue (₹)	5-year EBITDA (₹)	Payback
Conservative	8,000 vehicles/day, EV share ×0.7, stop rate ×0.8	19,03,322	-12,33,321	beyond 7 yrs
Base	12,000 vehicles/day	50,98,183	33,74,582	5.5 yrs
Optimistic	16,000 vehicles/day, EV share ×1.3, stop rate ×1.2	1,06,04,221	1,13,15,861	3.8 yrs

What this means. The project is viable only if traffic and EV adoption reach roughly the base case. A paid 7-day traffic count, a signed tariff from the distribution company and fleet tie-ups (taxis, intercity EV buses) are the three steps that most reduce risk. Solar on the canopy, a small café or shop, and a higher tariff for 120 kW charging are upside options not counted here.

8. Risks & next steps

Risk	Effect	Mitigation
Traffic lower than assumed	Slower payback	7-day count before work order; stage the 120 kW unit if counts are weak
Slow EV uptake in the district	Low sessions in years 1–2	Tie-ups with taxi and bus operators; listing on charging apps
Power cost above ₹7/kWh	Margin falls ₹1 per kWh for each ₹1 rise	Confirm EV tariff and demand charges in writing
Statutory delays (sanction, CEIG)	Opening date slips	Start applications now; fees at actuals
Road-access permission	No legal entry from the highway	Obtain NHAI/PWD permission for the entry cut

Next steps

1. Visit the site: measure plot depth, locate nearest HT pole, take soil samples.
2. Run a 7-day traffic count (by vehicle class) and update section 6.
3. Confirm the distribution company (HESCOM or BESCOM) and the EV tariff.
4. Sign the work order and pay the 50% advance (see attached invoice).
5. Apply for sanction and CEIG approval; order the Exicom chargers.
6. Install and commission in 30–45 days from advance and approvals.

Sources and basis: The Lynx quotation dated 06-10-2026; charger and canopy prices as provided by the client; Esri World Imagery; OpenStreetMap (road name and classification); the web sources linked in section 6. Demand and revenue figures are estimates by Sonar.EV.